

| mapping_id                                            | source_ontology_id | source_entity_type                                                   | source_iri                                                                                                                                | source_prefix | source_label           | target_ontology_id | target_entity_type                                                   | target_iri                                                                                                                                      | target_prefix | target_label               | alignment_relation                 | direction                                           | mapping_notes & justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|--------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------|--------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------|------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Stable mapping identifier, e.g., DPPO_CEO_0001</i> | DPP(DPPO)/CEON     | Class (ObjectProperty /DataProperty /AnnotationProperty /Individual) | Absolute IRI                                                                                                                              | source prefix | Human-readable label   | DPP(DPPO)/CEON     | Class (ObjectProperty /DataProperty /AnnotationProperty /Individual) | Absolute IRI                                                                                                                                    | target prefix | Human-readable label       | Controlled vocabulary (see README) | source_to_target / target_to_source / bidirectional | rationale, ignored by generator for logic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| DPPO_CEO_0001                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-odp/Product">http://w3id.org/dppo/ontology/dpp-odp/Product</a>                                 | dpp-odp       | Product                | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/product/Product">http://w3id.org/CEON/ontology/product/Product</a>                                       | ceon-product  | Product                    | subclass_of                        | target_to_source                                    | DPPO's Product is broad and is not disjoint from material, whereas CEON's Product covers physical products but excludes materials, which CEON places under ceon:Matter. Every ceon:Product is therefore a dppo:Product, but not every dppo:Product is a ceon:Product. By the subsumption criterion, the semantically DPPO defines Material as a subclass of Product, representing e.g. a bulk material or commodity that can itself have a passport. CEON's Material is a class under Matter, conceptually "the stuff products are made of." Despite the hierarchy difference, these classes refer to the same real-world category (materials like steel, plastic, wood). We align dppo:Material to ceon-material:Material as equivalent in intention, with the caveat that in CEON a Material is not a Product. In practice, this means if a DPP instance is typed as Material, we treat it as an instance of CEON Material (and not simultaneously a CEON Product). This alignment is semantically sound. DPP's broad notion of product covers materials, but when translating to CEON, we cast those to the Material branch. This is a specialization alignment, effectively partitioning DPP's Product category into CEON's Product vs Material depending on the instance's DPP subtype. |
| DPPO_CEO_0002                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-core/Material">http://w3id.org/dppo/ontology/dpp-core/Material</a>                             | dpp-core      | Material               | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/material/Material">http://w3id.org/CEON/ontology/material/Material</a>                                   | ceon-material | Material                   | equivalent_class                   | bidirectional                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DPPO_CEO_0003                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-core/Substance">http://w3id.org/dppo/ontology/dpp-core/Substance</a>                           | dpp-core      | Substance              | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/material/ChemicalSubstance">http://w3id.org/CEON/ontology/material/ChemicalSubstance</a>                 | ceon-material | ChemicalSubstance          | subclass_of                        | target_to_source                                    | DPPO uses Substance for basic chemical entities (elements or compounds) that might appear in a product's composition or as regulated substances. CEON explicitly defines ChemicalSubstance and ChemicalElement classes. This can be seen as a specialization of DPPO's Substance class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| DPPO_CEO_0004                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-comp/SubstanceOfConcern">http://w3id.org/dppo/ontology/dpp-comp/SubstanceOfConcern</a>         | dpp-comp      | SubstanceOfConcern     | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/material/ChemicalEntity">http://w3id.org/CEON/ontology/material/ChemicalEntity</a>                       | ceon-material | ChemicalEntity             | subclass_of                        | source_to_target                                    | CEON does not have built-in classes for "substance of concern," but DPPO defines these as special Substance subclasses per EU regulation. We align this by treating DPPO's concern subclasses as subclasses of CEON's ChemicalEntity or ChemicalSubstance. For example, map dppo:ToxicSubstance as subclass of ceon-material:ChemicalSubstance, and similarly for others. This maintains the intent (they are chemical substances) within CEON's structure. The alignment type is partial: it introduces new specialization in CEON's hierarchy. CEON's material module can accommodate this extension (it was designed to be extended for domain-specific material classes). The mapping is justified because it enriches CEON with DPPO's regulatory nuance – something CEON currently lacks. Conversely, from a CEON perspective, one could treat these alignments as annotations (e.g., link a substance to an external regulatory classification). But for ontology alignment, subclassing is straightforward and retains semantics (any DPP ToxicSubstance is a CEON ChemicalSubstance with an added property "is toxic" or simply by class membership).                                                                                                                               |
| DPPO_CEO_0005                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-odp/hasPart">http://w3id.org/dppo/ontology/dpp-odp/hasPart</a>                                 | dpp-odp       | hasPart                | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/product/hasProductComponent">http://w3id.org/CEON/ontology/product/hasProductComponent</a>               | ceon-product  | hasProductComponent        | subproperty_of                     | target_to_source                                    | DPPO uses one generic hasPart for products; CEON splits into model- and item-level shortcuts. Making both CEON properties subproperties of DPPO's hasPart preserves DPPO queries and CEON's extra precision. DPPO also infers partonomy from refined composition (see below).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DPPO_CEO_0006                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-odp/hasPart">http://w3id.org/dppo/ontology/dpp-odp/hasPart</a>                                 | dpp-odp       | hasPart                | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/product/hasProductObjectComponent">http://w3id.org/CEON/ontology/product/hasProductObjectComponent</a>   | ceon-product  | hasProductObjectComponent  | subproperty_of                     | target_to_source                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DPPO_CEO_0007                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-info/CompositionInformation">http://w3id.org/dppo/ontology/dpp-info/CompositionInformation</a> | dpp-info      | CompositionInformation | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/product/Composition">http://w3id.org/CEON/ontology/product/Composition</a>                               | ceon-product  | Composition                | equivalent_class                   | bidirectional                                       | Both classes rely composition. DPPO's CompositionInformation is a piece of information about the composition of a product. CEON's Composition represents a composing relation. Stated as an equivalence in the Material Composition section of the accompanying paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DPPO_CEO_0008                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-comp/Component">http://w3id.org/dppo/ontology/dpp-comp/Component</a>                           | dpp-core      | Component              | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/product/Product">http://w3id.org/CEON/ontology/product/Product</a>                                       | ceon-product  | Product                    | subclass_of                        | source_to_target                                    | CEON does not define a separate component class; components are products related through ceon-product:hasProductComponent. DPPO's Component, itself a subclass of dpp-odp:Product, is therefore aligned as a specialisation of ceon-product:Product. Stated in the Product Component and Partonomy Alignment section of the accompanying paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| DPPO_CEO_0009                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-comp/aboutWhole">http://w3id.org/dppo/ontology/dpp-comp/aboutWhole</a>                         | dpp-comp      | aboutWhole             | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/product/associatedWithProductModel">http://w3id.org/CEON/ontology/product/associatedWithProductModel</a> | ceon-product  | associatedWithProductModel | equivalent_property                | bidirectional                                       | Both properties anchor a refined composition to the whole it describes. Stated as an equivalence in the Product Component and Partonomy Alignment section of the accompanying paper for the product-to-product case. Note that in the published DPPO the domain of dpp-comp:aboutWhole is stated using a version-qualified IRI (dpp-info/0.1/CompositionInformation) that does not match the versioned class declared in dpp-info, while ceon-product:associatedWithProductModel is declared with domain ProductComposition, a specialisation of ceon-product:Composition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DPPO_CEO_0010                                         | DPPO               | Class                                                                | <a href="http://w3id.org/dppo/ontology/dpp-info/DPPInformation">http://w3id.org/dppo/ontology/dpp-info/DPPInformation</a>                 | dpp-info      | DPPInformation         | CEON               | Class                                                                | <a href="http://w3id.org/CEON/ontology/resource/ODPInformation">http://w3id.org/CEON/ontology/resource/ODPInformation</a>                       | resourceODP   | Information                | subclass_of                        | source_to_target                                    | DPPO's DPPInformation is a piece of information contained in a DPP and about a product; CEON's resourceODP:Information is the general notion of information about a resource. Stated in the Digital Product Passport as Document section of the accompanying paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DPPO_CEO_0011                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-info/containsInformation">http://w3id.org/dppo/ontology/dpp-info/containsInformation</a>       | dpp-info      | containsInformation    | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/resource/ODP/containsInformation">http://w3id.org/CEON/ontology/resource/ODP/containsInformation</a>     | resourceODP   | containsInformation        | subproperty_of                     | source_to_target                                    | DPPO relates a DPP to the pieces of information it contains; the CEON property is the general containment relation between a collection of information and its parts. Stated in the Digital Product Passport as Document section of the accompanying paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DPPO_CEO_0012                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-odp/describes">http://w3id.org/dppo/ontology/dpp-odp/describes</a>                             | dpp-odp       | describes              | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/resource/ODP/isAbout">http://w3id.org/CEON/ontology/resource/ODP/isAbout</a>                             | resourceODP   | isAbout                    | subproperty_of                     | source_to_target                                    | Expressed as a subsumption rather than an equivalence to prevent unintended inferences arising from the domain restriction of dpp-odp:describes. Stated in the Digital Product Passport as Document section of the accompanying paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DPPO_CEO_0013                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-info/isAbout">http://w3id.org/dppo/ontology/dpp-info/isAbout</a>                               | dpp-info      | isAbout                | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/resource/ODP/isAbout">http://w3id.org/CEON/ontology/resource/ODP/isAbout</a>                             | resourceODP   | isAbout                    | subproperty_of                     | source_to_target                                    | Generalises DPPO_CEO_0012: dpp-odp:describes is declared a subproperty of dpp-info:isAbout within DPPO, and both isAbout properties relate a piece of information to the entity it describes. Additional mapping, not discussed in the paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| DPPO_CEO_0014                                         | DPPO               | ObjectProperty                                                       | <a href="http://w3id.org/dppo/ontology/dpp-comp/aboutPart">http://w3id.org/dppo/ontology/dpp-comp/aboutPart</a>                           | dpp-comp      | aboutPart              | CEON               | ObjectProperty                                                       | <a href="http://w3id.org/CEON/ontology/product/compositionOf">http://w3id.org/CEON/ontology/product/compositionOf</a>                           | ceon-product  | compositionOf              | subproperty_of                     | source_to_target                                    | Counterpart of DPPO_CEO_0009 for the constituent side: both relate a refined composition to the entity that the composition is composed of. Additional mapping, not discussed in the paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |